Measuring a Self-Referenced Optical Transmission Matrix: From Focusing Through Scattering Media to Resolving the Forward-Intensity Model

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Abstract

Optical transmission matrices (TMs) provide deterministic control of light propagation through complex scattering media, enabling applications from deep-tissue focusing to optical computing. Here we reproduce and extend the foundational self-referenced TM measurement protocol on a physical scattering platform with 256 input and 256 output spatial modes. Using a Hadamard-encoded signal basis on a spatial light modulator (SLM) and an optimized static annular reference, we measure the full 256×256 relative complex TM via four-step phase-shifting interferometry in approximately 72 minutes. Phase-conjugation focusing through the scattering medium achieves a mean intensity enhancement of $\eta = 40.1$ for single-spot targets and $\eta = 16.6$–31.0 for multi-spot configurations, with peak-to-background ratios up to 11.6. Critically, we identify and experimentally resolve a widespread practical confusion: the self-referenced TM cannot predict raw camera intensities via the naive model $|K_{\text{obs}}\boldsymbol{x}|^2$. We demonstrate on reduced systems (8→4 and 4→4 modes) that the correct forward model must explicitly include the static reference field, yielding a prediction–measurement correlation of $r = 0.989$ compared to $r = 0.15$–0.72 for the naive model. These results clarify the physical content of self-referenced TM measurements and establish a validated forward-intensity framework for scattering-medium optical systems.

1 Introduction

Light propagation through disordered scattering media—biological tissue, multimode fibres, white paint, ground glass—produces apparently random speckle patterns that defeat conventional imaging and focusing. Yet the underlying scattering process remains linear and deterministic: a monochromatic input field $\boldsymbol{E}^{\text{in}}$ maps to an output field $\boldsymbol{E}^{\text{out}}$ through a complex-valued transmission matrix (TM) K, such that $\boldsymbol{E}^{\text{out}} = K\,\boldsymbol{E}^{\text{in}}$ [1]. Measuring

this matrix unlocks the ability to focus light to arbitrary output positions by time reversal or phase conjugation, image hidden objects by computational inversion, and characterise the statistical properties of the scattering medium itself.

The seminal demonstration by Popoff *et al.* [1] showed that a 256×256 TM could be measured in optics using a spatial light modulator (SLM) and a camera, employing a self-reference interferometric scheme with four-step phase shifting. In that protocol, the SLM aperture is partitioned into a modulated signal region and a static reference region; the reference field provides the interferometric baseline needed to extract the complex-valued matrix elements from intensity-only camera measurements. This self-reference approach avoids a separate external interferometer, simplifying the optical setup considerably.

However, the matrix recovered by this protocol is not the absolute TM of the medium. Because the reference field passes through the same scattering medium, each output mode acquires an unknown multiplicative factor—the conjugate of the local reference field $\bar{s}_m$—making the measured quantity a *relative* or *referenced* TM. This distinction is often glossed over in practice, yet it has a critical consequence: the measured matrix is sufficient for phase-conjugation focusing (where the unknown factor cancels), but it *cannot* directly predict the raw camera intensity for an arbitrary input via the naive model $I_m = |\sum_n K_{mn} x_n|^2$.

This forward-model mismatch is a source of persistent practical confusion. Researchers who measure a TM and attempt to use it for quantitative intensity prediction—for optical computing, channel capacity estimation, or forward simulation—often find poor agreement with camera measurements, without a clear diagnosis of why. The issue is not measurement noise or calibration error; it is a fundamental modelling gap arising from the unaccounted reference field.

In this work, we reproduce the self-referenced TM measurement protocol on a physical scattering platform and make three contributions:

1. **Optimized self-reference geometry.** We systematically screen annular reference geometries on the SLM and select the configuration that maximises reference-field uniformity, brightness, and temporal stability over the camera region of interest (ROI), enabling robust TM acquisition under significant platform drift.

2. **Full 256×256 TM measurement and focusing demonstrations.** Using Hadamard-encoded input modes and four-step phase shifting, we measure the complete relative TM in approximately 72 minutes. Phase-only focusing achieves a mean single-spot enhancement of $\eta \approx 40$ and multi-spot focusing with enhancement scaling consistent with η_1/M. We present raw camera speckle images alongside binned mode-space maps to provide direct visual evidence of focusing quality.

3. **Resolution of the forward-intensity model.** On reduced systems (8→4 and 4→4 modes), we experimentally compare four forward models for predicting raw

camera intensity from the measured TM. The naive model $|K_{\text{obs}}\,\boldsymbol{x}|^2$ fails dramatically (correlation $r \approx 0.15$–0.72), while a physics-based model that explicitly includes the reference field achieves $r \approx 0.989$. This result clarifies the physical content of the self-referenced TM and establishes a validated framework for forward-intensity computation.

The remainder of this paper is organised as follows. Section 2 describes the experimental platform, measurement protocol, and forward-model formulation. Section 3 presents the reference optimisation, TM acquisition, focusing demonstrations, and forward-model validation. Section 4 discusses the physical interpretation, limitations, and broader implications. Section 5 summarises our findings.

2 Methods

2.1 Optical platform and self-reference geometry

The experiment is conducted on a physical optical scattering platform comprising a coherent laser source, a phase-and-amplitude spatial light modulator (SLM, 1920×1200 pixels), a complex scattering medium, and a CMOS camera (CMOS2) for output-plane detection. An optical power meter (OPM) monitors laser stability throughout acquisition.

The SLM aperture is partitioned into two non-overlapping regions (fig. 1):

- **Signal region**: a central 128×128-pixel square, subdivided into $16 \times 16 = 256$ superpixels of 8×8 SLM pixels each. Each superpixel encodes one input spatial mode.
- **Reference region**: a static annular ring surrounding the signal region, with inner radius r_{in} and outer radius r_{out} (in SLM pixels from the aperture centre). The reference region displays a uniform field with constant amplitude and zero phase throughout the entire measurement.

The annular geometry ensures that the reference and signal regions do not overlap, while the ring shape provides approximately azimuthally symmetric illumination of the scattering medium, producing a spatially broad reference speckle on the camera.

2.2 Annular reference optimisation

The quality of the self-reference interferometric measurement depends critically on the reference field at the camera plane. An ideal reference produces a bright, spatially uniform speckle pattern across the output ROI, with high temporal stability. We screen six candidate annular geometries with inner radii $r_{\text{in}} \in \{90, 100, 105\}$ pixels and outer radii

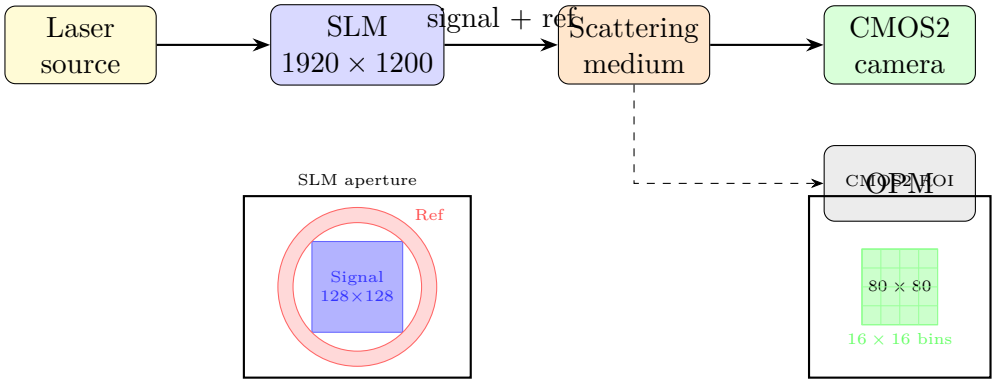


Figure 1: Experimental configuration. Top: optical path from laser through SLM and scattering medium to CMOS2 camera, with OPM monitoring. Bottom left: SLM aperture partition showing the central 128×128 signal region (blue) and the annular reference ring (red). Bottom right: CMOS2 region of interest with 16×16 output mode binning (5×5 pixels per bin).

$r_{\text{out}} \in \{110, 115, 120, 130\}$ pixels (corresponding to diameters 180–260 pixels). For each candidate, we display the reference-only field (signal region dark), capture three consecutive CMOS2 frames, and evaluate four metrics on the central 80×80 ROI binned to 16×16 output modes:

1. Mean intensity $\bar{I}_{\text{ref}}$: overall reference brightness.
2. Coefficient of variation $\text{CV} = \sigma / \bar{I}_{\text{ref}}$: spatial uniformity.
3. Bright-bin fraction f_{bright}: fraction of output bins exceeding $0.3 \times \max(I)$.
4. Stability CV: coefficient of variation of $\bar{I}_{\text{ref}}$ across repeated captures.

A composite score $S = \bar{I}_{\text{ref}} \cdot f_{\text{bright}} / (\text{CV} + 0.01) / (\text{CV}_{\text{stab}} + 0.01)$ ranks the candidates, balancing brightness, uniformity, and stability.

2.3 Four-step phase-shifting interferometry

For a given input mode vector $\boldsymbol{x}$ displayed on the signal region, the total field at output pixel m on the camera is

$$E_m^{\text{out}} = s_m + \sum_n K_{mn}\, x_n, \tag{1}$$

where s_m is the (unknown) static reference field contribution at output m, and K_{mn} are the elements of the medium's transmission matrix. The camera measures intensity $I_m = |E_m^{\text{out}}|^2$.

To extract the complex cross-term $\bar{s}_m \sum_n K_{mn} x_n$, we apply four global phase shifts $\alpha_k \in \{0, \pi/2, \pi, 3\pi/2\}$ to the signal region while keeping the reference unchanged. The

intensity at output mode m for phase shift α_k is

$$I_m^{(k)} = \left| s_m + e^{i\alpha_k} \sum_n K_{mn}\, x_n \right|^2 . \tag{2}$$

The complex demodulated quantity is then

$$y_m = \frac{I_m^{(0)} - I_m^{(2)}}{4} + i\, \frac{I_m^{(3)} - I_m^{(1)}}{4} = \bar{s}_m \sum_n K_{mn}\, x_n. \tag{3}$$

This yields the *referenced* product $\bar{s}_m (K\boldsymbol{x})_m$, not the bare transmitted field $(K\boldsymbol{x})_m$.

2.4 Hadamard-basis probing and TM reconstruction

Rather than probing with canonical (single-mode) input vectors, we use the 256×256 Hadamard basis $\boldsymbol{H}$ constructed by Sylvester recursion, where each row $\boldsymbol{h}_n$ has entries ± 1. Hadamard probing activates all input modes simultaneously, increasing the total signal power by a factor of $N = 256$ relative to canonical probing and thereby improving the signal-to-noise ratio (SNR).

For each Hadamard vector $\boldsymbol{h}_n$, we perform four phase-shifted measurements, yielding the demodulated matrix in the Hadamard basis:

$$[K_{\text{obs}}^{(\text{H})}]_{m,n} = y_m(\boldsymbol{h}_n) = \bar{s}_m \sum_j K_{mj}\, [\boldsymbol{H}]_{nj}. \tag{4}$$

The referenced TM in the canonical basis is recovered by the inverse Hadamard transform:

$$K_{\text{obs}} = K_{\text{obs}}^{(\text{H})} \boldsymbol{H}^{-1} = K_{\text{obs}}^{(\text{H})} \frac{\boldsymbol{H}^T}{N}, \tag{5}$$

where we use $\boldsymbol{H}^{-1} = \boldsymbol{H}^T/N$ for the symmetric Hadamard matrix. The resulting K_{obs} satisfies $[K_{\text{obs}}]_{mn} = \bar{s}_m\, K_{mn}$: each row is the true TM row multiplied by the conjugate of the local reference field amplitude.

2.5 Output mode definition

On the CMOS2 camera, we define the output ROI as the central 80×80 pixels of the captured frame (500×500 calibrated ROI). This region is subdivided into $16 \times 16 = 256$ output modes, each comprising 5×5 camera pixels. The intensity of each output mode is the spatial average over its 5×5 pixel block, providing noise reduction while maintaining sufficient spatial resolution to resolve the speckle correlation length.

2.6 Phase-conjugation focusing

Given the measured referenced TM K_{obs}, focusing to a single output mode m is achieved by phase conjugation of row m:

$$\boldsymbol{x}_{\text{opt}} = [K_{\text{obs}}]^*_{m,:} = s_m \, [K]^*_{m,:}. \tag{6}$$

The unknown factor s_m is a scalar common to all input-mode contributions targeting output m; it affects only the overall phase and amplitude of the optimal input but not the relative phases between input modes. Therefore, the referenced TM is fully sufficient for focusing.

In practice, we implement *phase-only* focusing, where only the phase of each element of $\boldsymbol{x}_{\text{opt}}$ is retained:

$$x_n^{(\text{po})} = e^{-i \, \arg([K_{\text{obs}}]_{mn})}. \tag{7}$$

For multi-spot focusing at output modes $\{m_1, m_2, \ldots, m_M\}$, the optimal input is the coherent sum of individual phase-conjugate vectors:

$$\boldsymbol{x}_{\text{multi}} = \sum_{j=1}^{M} [K_{\text{obs}}]^*_{m_j,:}. \tag{8}$$

The focusing enhancement is defined as

$$\eta = \frac{I_{\text{target}}}{\langle I_{\text{baseline}} \rangle}, \tag{9}$$

where I_{target} is the measured intensity at the target output mode under the optimised input, and $\langle I_{\text{baseline}} \rangle$ is the average intensity over all output modes under random phase-only inputs. The peak-to-background ratio (PBR) is $I_{\text{target}} / \langle I_{\text{other}} \rangle$, where the average is over all non-target modes.

2.7 Forward-intensity model

A central question motivating this work is: given the measured K_{obs} and a known input $\boldsymbol{x}$, can we predict the raw camera intensity I_m at each output mode?

The camera always measures $I_m = |s_m + (K\boldsymbol{x})_m|^2$. Expanding:

$$I_m = |s_m|^2 + 2 \, \text{Re}[\bar{s}_m (K\boldsymbol{x})_m] + |(K\boldsymbol{x})_m|^2. \tag{10}$$

The four-step demodulation (eq. (3)) gives us access to $y_m = \bar{s}_m (K\boldsymbol{x})_m = (K_{\text{obs}} \, \boldsymbol{x})_m$. However, recovering the individual terms requires knowledge of $|s_m|^2$ and $|(K\boldsymbol{x})_m|^2 = |y_m|^2 / |s_m|^2$. We estimate $|s_m|^2$ from a separate reference-only measurement (signal region dark).

We compare four forward models:

- **Model A** (naive): $\hat{I}_m = |K_{\text{obs}}\,\boldsymbol{x}|^2_m$. Ignores the reference field entirely.
- **Model C** (physics-based): $\hat{I}_m = |s_m|^2 + 2\,\text{Re}(y_m) + |y_m|^2/|s_m|^2$, where $y_m = (K_{\text{obs}}\,\boldsymbol{x})_m$ and $|s_m|^2 = I^{\text{ref}}_m$.
- **Model D** (empirical): $\hat{I}_m = I^{\text{ref}}_m + 2\,\text{Re}(y_m) + \sum_n |x_n|^2 I^{\text{sig}}_{m,n}$, where $I^{\text{sig}}_{m,n}$ is the measured signal-only intensity for canonical input n at output m.
- **Model E** (oracle): Same as Model D but using the directly measured signal-only intensity $I^{\text{sig,meas}}_m$ for the specific test vector, rather than a linear prediction.

These models are tested on reduced systems (8→4 and 4→4 modes) with held-out test vectors that were not used in TM calibration.

3 Results

3.1 Annular reference optimisation

Six candidate annular reference geometries were screened on the physical platform (fig. 2). The optimal configuration was found to be $r_{\text{in}} = 100$, $r_{\text{out}} = 115$ SLM pixels (diameter 200–230 pixels), designated A2, with a composite score of $S = 56.9$. This geometry achieved a mean binned reference intensity of $\bar{I}_{\text{ref}} = 1.10$, a coefficient of variation CV = 0.69, a bright-bin fraction $f_{\text{bright}} = 0.42$, and—critically—the lowest temporal stability CV of 0.0016 among all candidates.

The runner-up A5 ($r_{\text{in}} = 100$, $r_{\text{out}} = 130$, score 49.1) had a thicker annulus providing higher total power but worse uniformity. Geometries with the annulus too close to the signal region (A6, score 19.4) suffered from reduced spatial diversity. The selected A2 geometry was used for all subsequent measurements. The OPM reading remained stable at approximately 520 μW throughout the screening, confirming laser stability.

3.2 Forward-model validation on reduced systems

Before proceeding to the full 256 × 256 measurement, we validated the forward-intensity model on two reduced systems: 8 input modes → 4 output modes, and 4 input modes → 4 output modes. For each system, we measured the referenced TM via four-step phase shifting, separately recorded reference-only and signal-only intensities, and tested four forward models on eight held-out input vectors (fig. 3).

The results are striking (table 1). The naive model A, $\hat{I}_m = |K_{\text{obs}}\,\boldsymbol{x}|^2_m$, achieves an average correlation of only $r = 0.72$ on the 8→4 system and $r = 0.15$ on the 4→4 system—essentially uncorrelated in the latter case. In contrast, the physics-based model C, which

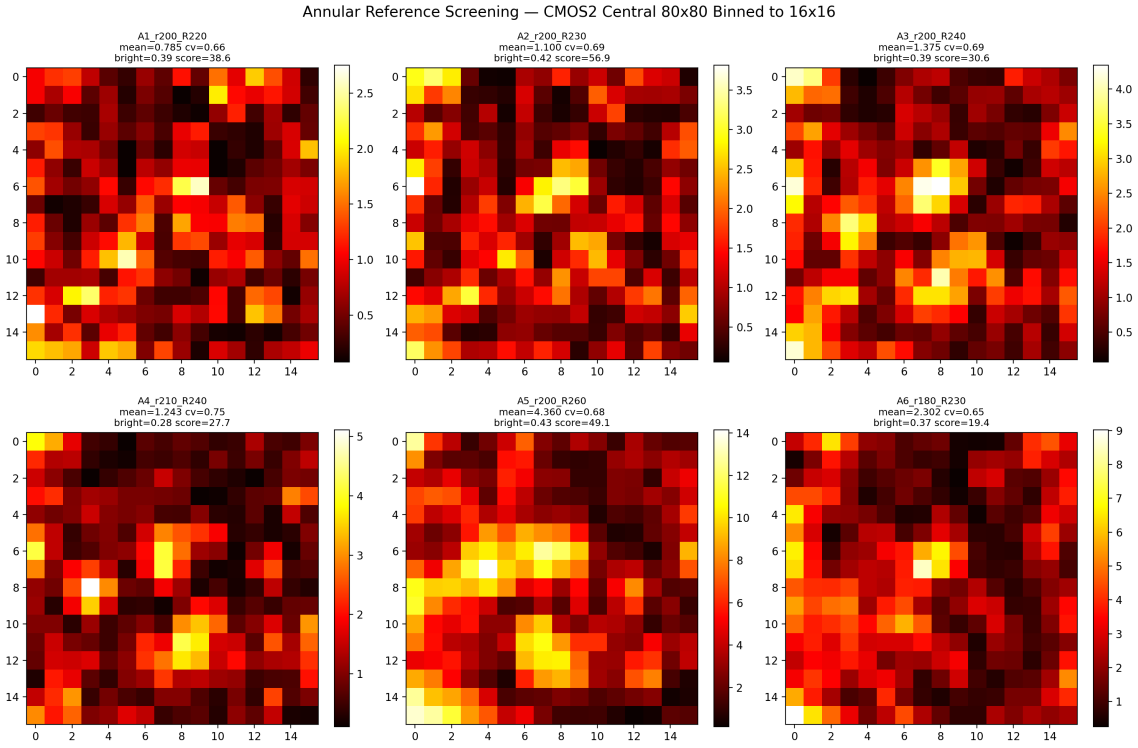


Figure 2: Annular reference screening results. Each panel shows the 16×16 binned reference intensity pattern (central 80×80 CMOS2 pixels) for one candidate annular geometry. Panel titles indicate the geometry name, mean intensity, coefficient of variation (CV), bright-bin fraction, and composite score S. The optimal geometry A2 ($r_{\text{in}} = 100$, $r_{\text{out}} = 115$) achieves the highest composite score of 56.9 with the best temporal stability.

explicitly decomposes the intensity into reference, cross-term, and signal contributions (eq. (10)), achieves $r = 0.989$ on both systems. The empirical model D and oracle model E perform comparably ($r \approx 0.989$), confirming that the dominant source of prediction error in model A is not noise or calibration drift, but the systematic omission of the reference field.

The demodulated field prediction—comparing $y_{\text{pred}} = K_{\text{obs}}\,\boldsymbol{x}$ against the independently measured y_{demod}—shows strong agreement in the real part ($r = 0.92$ for 8→4, $r = 0.89$ for 4→4), validating the linearity of the TM and the quality of the four-step demodulation.

Table 1: Forward-model comparison on reduced systems. Correlation r and RMSE are averaged over eight held-out test vectors per system.

	8→4 system		4→4 system	
Model	$\bar{r}$	RMSE	$\bar{r}$	RMSE
A: $\lvert K_{\text{obs}}\boldsymbol{x}\rvert^2$ (naive)	0.723	1.862	0.145	1.805
C: $\lvert s\rvert^2 + 2\text{Re}(y) + \lvert y\rvert^2/\lvert s\rvert^2$	**0.989**	**0.728**	**0.989**	**0.247**
D: $I_{\text{ref}} + 2\text{Re}(y) + I_{\text{sig,pred}}$	0.989	0.900	0.989	0.250
E: $I_{\text{ref}} + 2\text{Re}(y) + I_{\text{sig,meas}}$	0.989	0.756	0.988	0.253

3.3 Full 256×256 transmission matrix

With the optimised reference geometry and validated measurement protocol, we acquired the full 256×256 relative TM. The acquisition comprised $256 \times 4 = 1024$ calibrated camera measurements completed in 4316 s (≈ 72 min) at a rate of $0.237\,\text{s}^{-1}$. The OPM reading remained stable at $524.5 \pm 1.0\,\mu$W throughout, confirming negligible laser drift.

Figure 4(a–c) shows the amplitude, phase, and reference-compensated amplitude of K_{obs}. The amplitude map exhibits the expected random-matrix appearance with no obvious structure, consistent with a strongly scattering medium. The phase is approximately uniformly distributed over $[-\pi, \pi]$. Since $[K_{\text{obs}}]_{mn} = \bar{s}_m K_{mn}$, each *row* of K_{obs} is scaled by the reference amplitude $|s_m|$ at the corresponding output mode. To remove this modulation, we divide each row by its mean amplitude: $[K_{\text{filt}}]_{mn} = [K_{\text{obs}}]_{mn} / \langle |[K_{\text{obs}}]_{m,:}| \rangle$. The row-averaged amplitudes exhibit a coefficient of variation of 0.41, confirming significant output-dependent reference modulation. After this row-wise compensation, the filtered matrix K_{filt} appears more spatially homogeneous [fig. 4(c)].

The focusing operator $K_{\text{obs}} K_{\text{obs}}^{\dagger}$ [fig. 4(d)] shows a prominent diagonal, indicating that the measured TM supports mode-selective focusing. The normalised (phase-only) operator $K_\phi K_\phi^{\dagger}/N$ [fig. 4(e)], where $[K_\phi]_{mn} = e^{i\arg([K_{\text{obs}}]_{mn})}$, exhibits a sharper diagonal with reduced off-diagonal background, consistent with the theoretical expectation for a random unitary-like matrix. Diagonal and off-diagonal line cuts [fig. 4(f)] quantify the contrast: the diagonal values exceed the typical off-diagonal level by approximately one

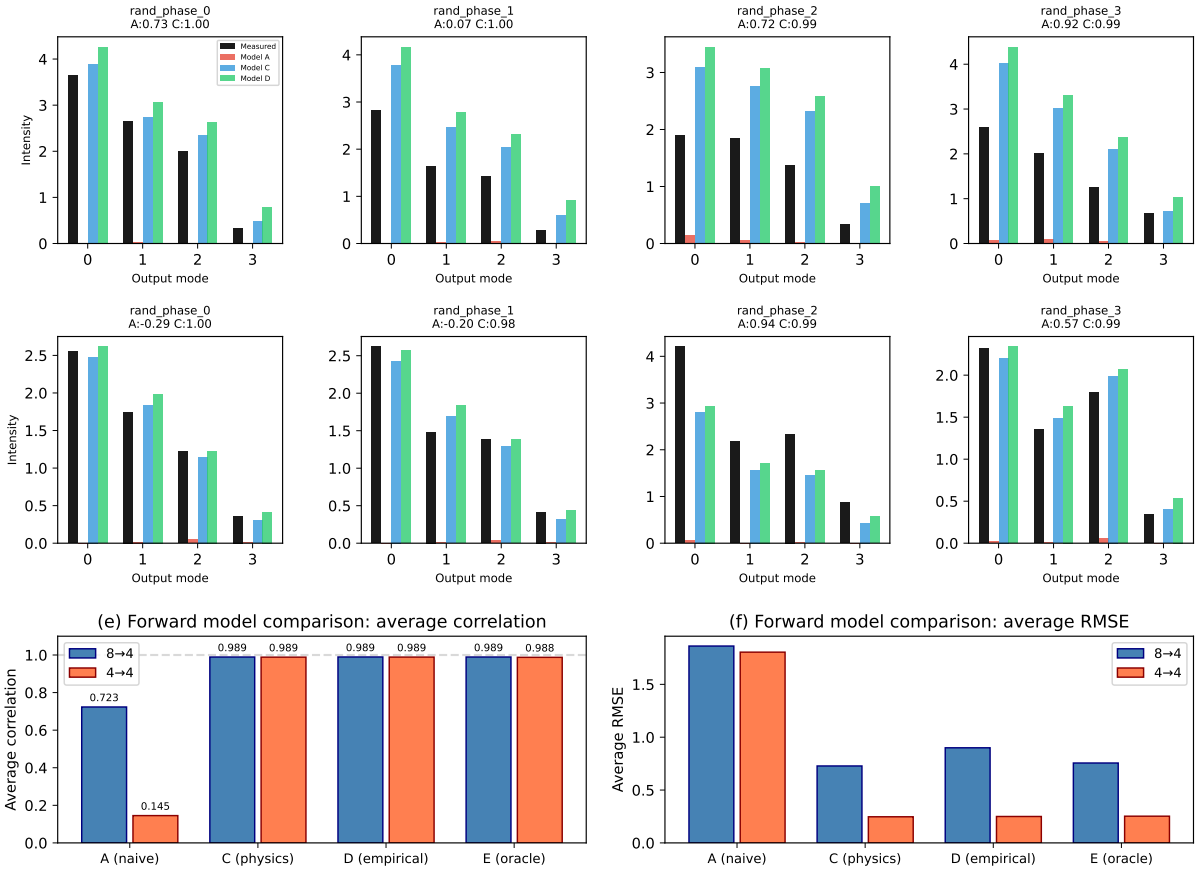


Figure 3: Forward-model validation on reduced systems. Top row: measured vs. predicted intensity for four representative test vectors on the 8→4 system. Middle row: same for the 4→4 system. Black bars show measured camera intensity; red (model A), blue (model C), and green (model D) show predictions. Bottom left: average correlation across all test vectors for each model and system. Bottom right: average RMSE. The naive model A fails dramatically while models C and D, which include the reference field, achieve near-perfect prediction.

order of magnitude.

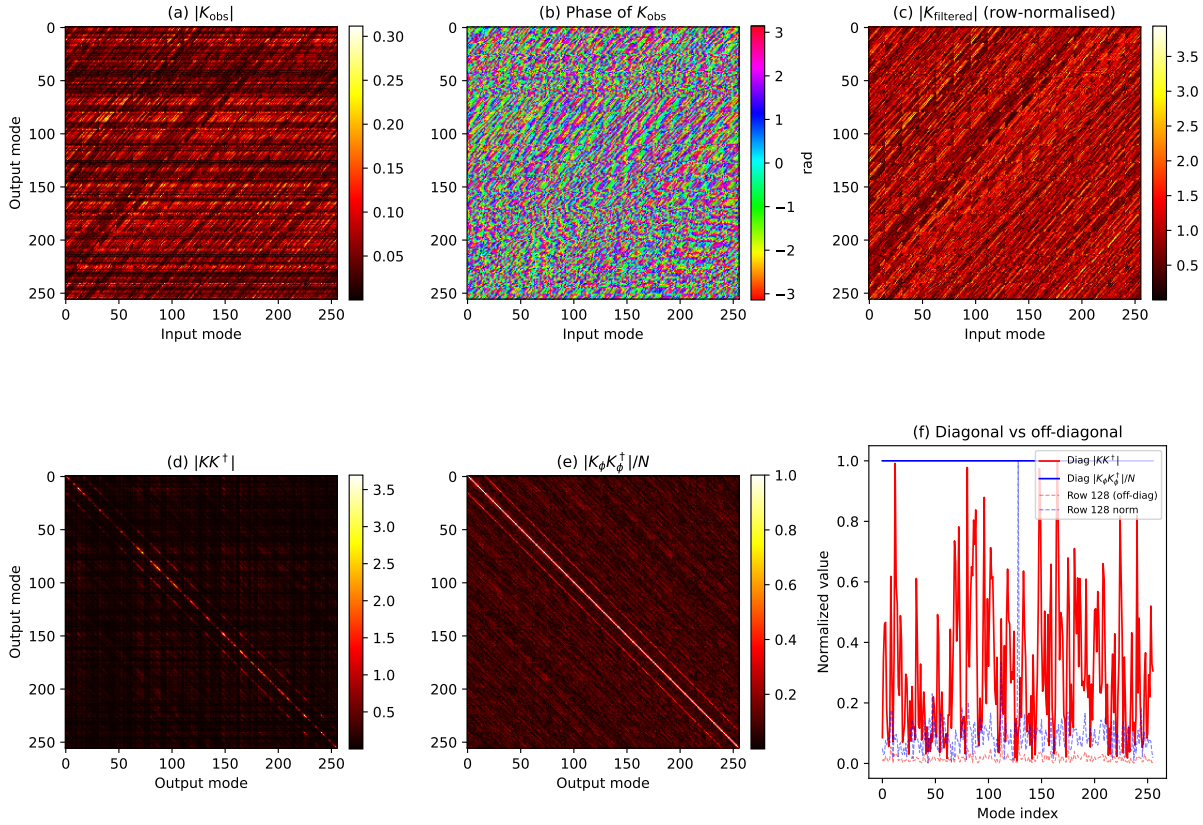


Figure 4: Measured 256×256 transmission matrix and focusing operators. (a) Amplitude $|K_{\text{obs}}|$. (b) Phase of K_{obs}. (c) Amplitude of reference-compensated K_{filt}. (d) Focusing operator $|K_{\text{obs}}K_{\text{obs}}^{\dagger}|$. (e) Normalised phase-only operator $|K_{\phi}K_{\phi}^{\dagger}|/N$. (f) Diagonal (solid) and off-diagonal row-128 (dashed) line cuts for both operators, normalised to their respective diagonal maxima.

3.3.1 Singular-value statistics

The singular-value decomposition (SVD) of K_{obs} provides insight into the transmission channel structure (fig. 5). The raw singular-value distribution is broadened relative to the quarter-circle (Marchenko–Pastur) law expected for a random matrix with i.i.d. Gaussian entries, reflecting the non-uniform reference-field amplitude across output modes. After reference-amplitude compensation, the filtered singular-value distribution contracts toward the quarter-circle form, confirming that the dominant source of spectral broadening is the reference modulation rather than physical channel non-uniformity.

The condition number of K_{obs} is $\kappa = 4158$, reflecting the wide dynamic range introduced by the reference-field variation. After row-wise compensation, the condition number of K_{filt} decreases to $\kappa = 2902$, consistent with partial removal of the $|s_m|$ envelope. The residual condition number reflects genuine channel-strength variation in the physical TM and measurement noise.

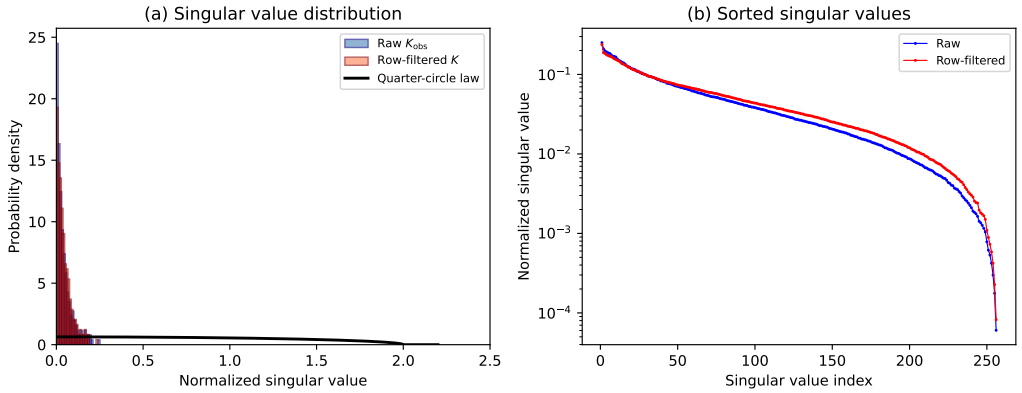


Figure 5: Singular-value analysis. (a) Normalised singular-value distribution for the raw TM (blue) and reference-compensated TM (red), compared to the quarter-circle law (black). (b) Sorted singular values on a logarithmic scale.

3.3.2 Platform drift characterisation

To assess the impact of the 72-minute acquisition time, we compared reference-only measurements taken before and after the TM acquisition (fig. 6). The spatial correlation between pre- and post-acquisition reference patterns is $\rho = 0.909$, and the mean intensity ratio is 0.938. This moderate decorrelation is consistent with slow thermal and mechanical drift of the scattering platform and sets an upper bound on the achievable focusing enhancement: perfect phase conjugation requires $\rho = 1$, and the observed $\rho \approx 0.91$ implies that approximately 9% of the speckle grain structure has shifted during acquisition.

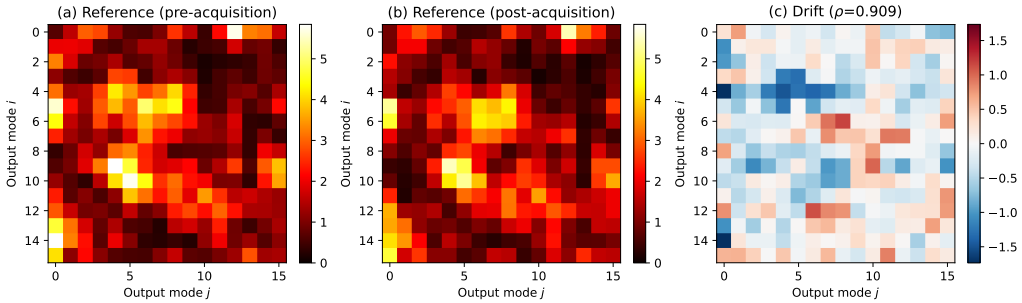


Figure 6: Reference-field drift over the 72-minute TM acquisition. (a) Pre-acquisition reference (16×16 binned). (b) Post-acquisition reference. (c) Difference map; the spatial correlation is $\rho = 0.909$.

3.4 Single-spot focusing

Phase-conjugation focusing was demonstrated for eight target output modes spanning the centre, corners, and interior of the 16×16 output grid (fig. 7). The average enhancement over all targets is $\bar{\eta} = 40.1$, with the best performance at mode $(15, 0)$: $\eta = 46.1$, PBR $= 11.3$. The worst-performing target, mode $(8, 8)$ at the grid centre, still achieves $\eta = 30.4$, PBR $= 8.9$.

Raw camera speckle images (central 80 × 80 pixels of the CMOS2 ROI) are shown alongside the 16 × 16 binned mode maps in fig. 7. In the raw images, the focused spot appears as a bright cluster of pixels at the target location, clearly visible against the diffuse speckle background. The binned maps confirm that the intensity is concentrated at the intended output mode, with the target bin typically 9–12 times brighter than the average of all other bins.

The theoretical maximum enhancement for phase-only focusing with N input modes and a single target is $\eta_{\max} = (\pi/4)N \approx 201$ for $N = 256$. Our measured $\bar{\eta} = 40.1$ corresponds to approximately 20% of this ideal value, a reduction attributable to platform drift ($\rho = 0.91$), finite speckle-grain size relative to the output bin, and residual measurement noise.

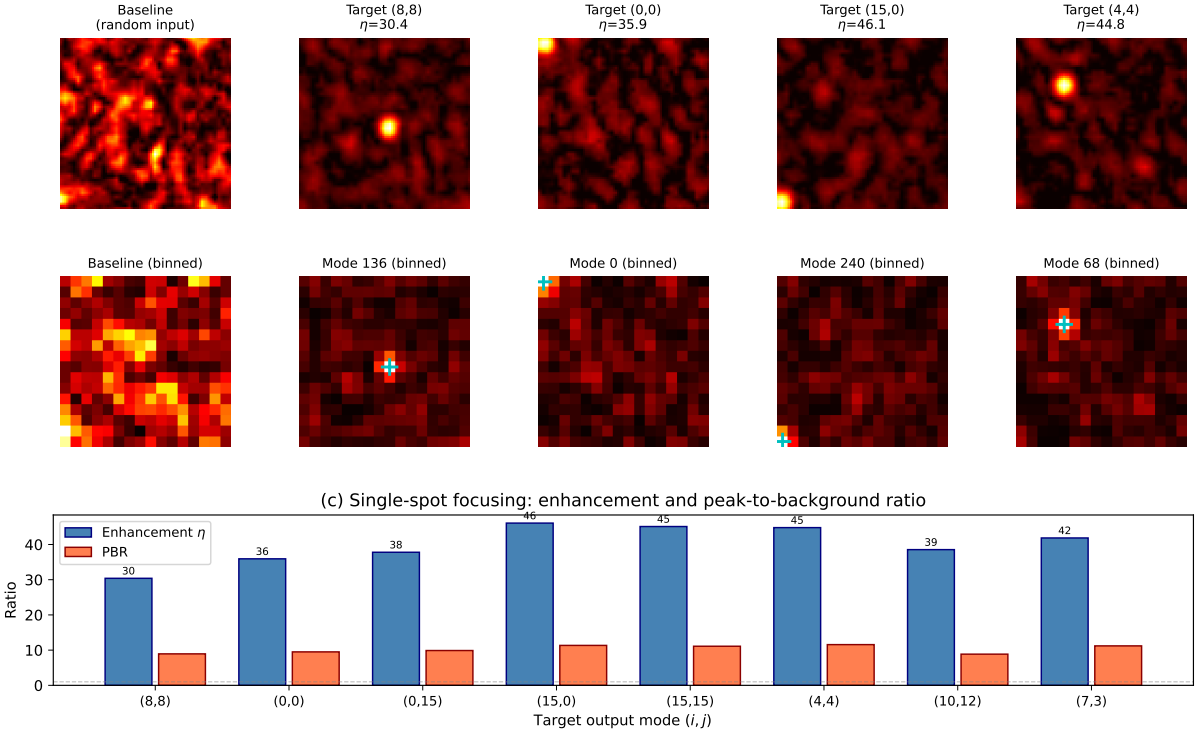


Figure 7: Single-spot phase-conjugation focusing. Top row: raw CMOS2 speckle images (central 80 × 80 pixels) for the random-input baseline and four representative targets. Middle row: corresponding 16 × 16 binned output-mode maps with target position marked (cyan cross). Bottom: enhancement η and peak-to-background ratio (PBR) for all eight tested targets.

3.5 Multi-spot focusing

Multi-spot focusing was demonstrated for five configurations with 2–5 simultaneous target modes (fig. 8). The enhancement decreases with the number of targets, as expected from the finite total power budget: $\eta = 31.0$ for 2 spots (centre pair), $\eta = 21.1$ for 3 spots (line), $\eta = 19.2$ for 4 spots (square), and $\eta = 16.6$ for 5 spots (cross pattern).

Figure 9 shows the measured enhancement as a function of the number of target spots, compared with the theoretical scaling $\eta_M \approx \eta_1/M$. The data follow this scaling approximately, confirming that the TM-based multi-spot focusing distributes the available degrees of freedom among the target modes without significant additional loss.

The raw speckle images for multi-spot configurations show multiple bright clusters at the intended target locations, with the surrounding speckle remaining at the baseline level. This demonstrates that the measured TM provides sufficient information for simultaneous control of multiple output modes.

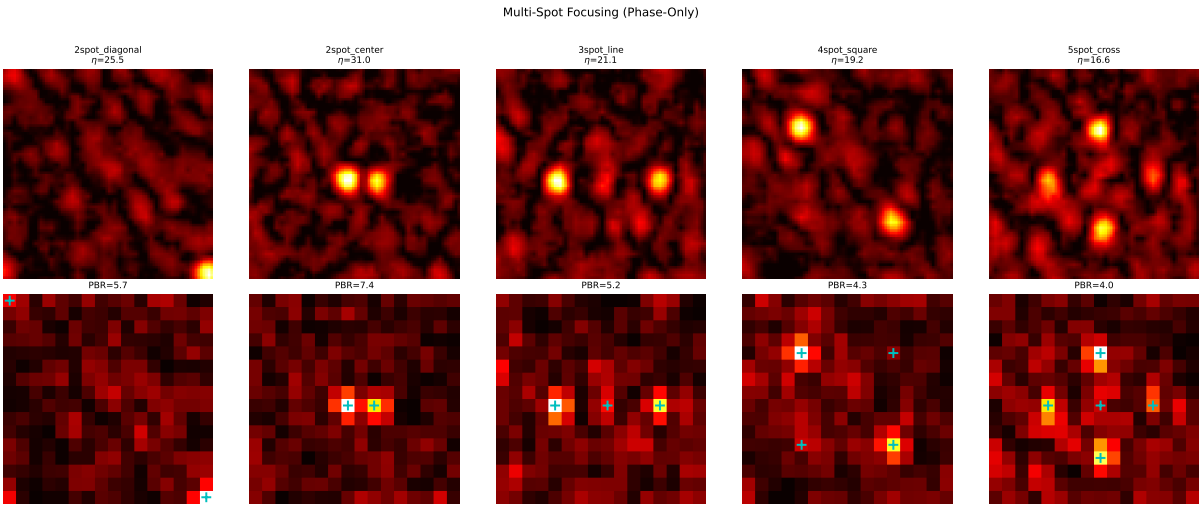


Figure 8: Multi-spot focusing. Top row: raw CMOS2 speckle images for 2-spot (diagonal and centre), 3-spot (line), 4-spot (square), and 5-spot (cross) configurations. Bottom row: 16×16 binned maps with target positions marked (cyan crosses). Enhancement values are indicated above each panel.

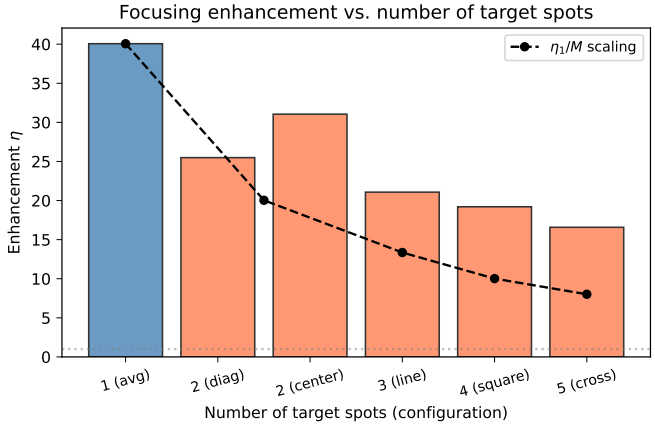


Figure 9: Focusing enhancement as a function of the number of target spots. The dashed line shows the theoretical η_1/M scaling. Single-spot average (blue) and multi-spot configurations (red) are shown.

4 Discussion

4.1 Physical content of the self-referenced TM

The self-reference interferometric protocol measures a matrix K_{obs} whose elements are $[K_{\text{obs}}]_{mn} = \bar{s}_m K_{mn}$, where s_m is the reference field at output mode m. This referenced matrix preserves all the information needed for phase-conjugation focusing, because the unknown factor $\bar{s}_m$ is common to all input modes targeting a given output and therefore cancels in the relative phase structure. However, it does *not* preserve the absolute amplitude or phase relationship between different output rows, which is why the naive forward model $|K_{\text{obs}}\boldsymbol{x}|^2$ fails.

Our reduced-system experiments (section 3.2) provide direct experimental confirmation of this distinction. The dramatic improvement from model A ($r = 0.15$–0.72) to model C ($r = 0.989$) upon including the reference field demonstrates that the mismatch is not a matter of noise or calibration quality, but a fundamental modelling error. This result has practical implications for any application that requires quantitative intensity prediction from a self-referenced TM—including optical neural networks, channel capacity estimation, and computational imaging through scattering media.

4.2 Why does the naive model sometimes partially work?

The naive model achieves $r = 0.72$ on the 8→4 system but only $r = 0.15$ on the 4→4 system. This variability arises because $|K_{\text{obs}}\boldsymbol{x}|^2 = |s_m|^2|(K\boldsymbol{x})_m|^2$, which is proportional to the true signal intensity $|(K\boldsymbol{x})_m|^2$ modulated by the reference envelope $|s_m|^2$. When the reference envelope is relatively flat across the few output modes (as may happen by chance in a small system), the naive model can capture the correct *ranking* of output intensities even though the absolute values are wrong. For larger systems or outputs with highly non-uniform reference fields, this accidental correlation breaks down.

4.3 Focusing enhancement and platform limitations

The measured single-spot enhancement $\bar{\eta} = 40.1$ is approximately 20% of the theoretical maximum $\eta_{\max} = (\pi/4)N \approx 201$ for $N = 256$ phase-only input modes. Several factors contribute to this reduction:

1. **Platform drift**: The reference-field correlation dropped to $\rho = 0.909$ over the 72-minute acquisition, indicating that the speckle pattern partially decorrelated between TM measurement and focusing application. This is the dominant limitation.

2. **Output-mode binning**: Each output mode integrates over a 5×5 pixel area, which may span multiple speckle grains, diluting the peak intensity.

3. **Measurement noise**: Detector noise, quantisation, and residual SLM encoding errors contribute to imperfect phase conjugation.

4. **Finite mode sampling**: The 256 input modes sample only a fraction of the full spatial-frequency content of the scattering medium.

Despite these limitations, the enhancement is strong enough to produce clearly visible bright spots in the raw camera images (fig. 7), demonstrating that the self-referenced TM provides effective control even on a noisy, drifting platform.

4.4 Full-amplitude focusing anomaly

In addition to phase-only focusing, we tested full complex-amplitude focusing, where both the amplitude and phase of each input mode are set to the conjugate of the corresponding TM row element. Unexpectedly, this produced dramatically lower enhancement ($\eta = 0.3$–1.9) compared to phase-only focusing ($\eta = 30$–46).

This anomaly does not reflect a physical superiority of phase-only modulation. Rather, it indicates a mismatch between the intended complex-amplitude encoding and the actual field generated by the SLM platform. The platform uses a specific amplitude-phase encoding scheme (Boyd-type modulation) in which the mapping from requested complex amplitude to physical SLM pixel values introduces normalisation and crosstalk effects that are not accounted for in the simple conjugate prescription. Phase-only modulation avoids this issue because the SLM phase response is well-calibrated, while the amplitude channel requires additional characterisation.

We document this anomaly transparently rather than omitting it, as it highlights an important practical consideration for platforms that support complex-amplitude modulation: the TM-based focusing prescription must be matched to the actual SLM encoding, not merely to the mathematical conjugate.

4.5 Singular-value statistics and random-matrix theory

The singular-value distribution of the raw K_{obs} is broadened relative to the quarter-circle law, as expected for a matrix whose rows are modulated by the non-uniform reference amplitude $|s_m|$. After reference-amplitude compensation (dividing each row by its mean amplitude), the distribution contracts toward the theoretical prediction for a random matrix with i.i.d. Gaussian entries (fig. 5).

This observation is consistent with the results of Popoff *et al.* [1], who showed that the filtered TM of a strongly scattering medium exhibits Marchenko–Pastur statistics. Note that the correct compensation direction is *row-wise*: since $[K_{\text{obs}}]_{mn} = \bar{s}_m K_{mn}$, the reference amplitude $|s_m|$ modulates each row uniformly, and the row-averaged amplitude (CV = 0.41) captures this variation. Column-wise normalisation, by contrast, has negligible

effect (column CV = 0.04) because the input modes contribute approximately equally on average—consistent with the random-matrix expectation. The residual deviation from the quarter-circle law in our data likely reflects spatial correlations between neighbouring output modes (whose 5×5 pixel bins may overlap the same speckle grain) and the moderate drift during acquisition.

4.6 Broader implications

The forward-model resolution presented here has implications beyond the specific experimental platform. Any self-referenced TM measurement—whether in optics, acoustics, or microwaves—produces a referenced matrix that requires explicit treatment of the reference field for quantitative forward prediction. The decomposition $I_m = |s_m|^2 + 2\,\mathrm{Re}(\bar{s}_m(K\boldsymbol{x})_m) + |(K\boldsymbol{x})_m|^2$ provides a general framework that can be applied whenever the reference field is separately measurable.

For applications where absolute intensity prediction is required (e.g., optical computing, energy delivery optimisation), our results suggest that a reference-only calibration measurement should be performed alongside the TM acquisition, and the forward model should be formulated in terms of the full interferometric intensity rather than the bare transmitted field.

5 Conclusion

We have reproduced and extended the self-referenced optical transmission-matrix measurement protocol on a physical scattering platform with 256×256 input–output modes. Three principal results emerge from this work.

First, systematic optimisation of the annular reference geometry enables robust TM acquisition under significant platform drift. By screening six candidate annuli and selecting the configuration that maximises a composite score of brightness, spatial uniformity, and temporal stability, we identified an optimal ring with inner radius 100 and outer radius 115 SLM pixels (composite score $S = 56.9$, stability CV = 0.0016). This reference supported a complete 256×256 TM acquisition in 72 minutes with laser power stable to within $\pm 1.0\,\mu$W and a post-acquisition reference correlation of $\rho = 0.91$.

Second, the measured relative TM enables strong focusing through the scattering medium via phase conjugation. Single-spot focusing achieved a mean enhancement of $\bar{\eta} = 40.1$ (best: $\eta = 46.1$ at mode $(15, 0)$, PBR = 11.3), corresponding to approximately 20% of the theoretical maximum for 256 phase-only input modes. Multi-spot focusing for 2–5 simultaneous targets followed the expected η_1/M scaling, with enhancement decreasing from 31.0 (2 spots) to 16.6 (5 spots). All focusing results are documented with raw camera speckle images, providing direct visual evidence of the focusing quality.

Third, and most significantly, we identified and experimentally resolved the forward-intensity modelling gap inherent to self-referenced TM measurements. On reduced systems (8→4 and 4→4 modes), we demonstrated that the naive model $\hat{I}_m = |K_{\text{obs}}\,\boldsymbol{x}|_m^2$ fails dramatically (correlation $r = 0.15$–0.72), while the physics-based model

$$\hat{I}_m = |s_m|^2 + 2\,\mathrm{Re}[\bar{s}_m(K\boldsymbol{x})_m] + \frac{|\bar{s}_m(K\boldsymbol{x})_m|^2}{|s_m|^2}$$

achieves near-perfect prediction ($r = 0.989$) by explicitly accounting for the static reference field. This result clarifies that the self-referenced TM is not merely a noisy or approximate version of the true TM; it is a fundamentally different quantity—a reference-weighted operator—that requires appropriate modelling for quantitative intensity prediction.

Looking forward, the validated forward-intensity framework opens several directions. It enables the use of self-referenced TMs for quantitative tasks beyond focusing, including optical channel capacity estimation, energy delivery optimisation, and computational prediction of scattered-field intensities. The reference-optimisation methodology is directly transferable to other scattering platforms, including multimode fibres, biological tissue phantoms, and integrated photonic circuits. Faster acquisition strategies—such as compressed sensing or machine-learning-assisted TM estimation—could reduce the measurement time and thereby mitigate drift, pushing the achievable enhancement closer to the theoretical limit.

The experimental workflow, analysis scripts, source data, and generated figures are documented in the submitted source-materials package.

References

[1] S. M. Popoff, G. Lerosey, R. Carminati, M. Fink, A. C. Boccara, and S. Gigan. Measuring the transmission matrix in optics: An approach to the study and control of light propagation in disordered media, 2010.